# NBTS for NEET-2024(XII Studying)_NBTS-05

1. For a uniformly charged ring of diameter $R$, the electric field on its axis has the maximum magnitude at a distance $x$ along axis of ring from its centre. The value of $x$ is

(a) $\frac{R}{2\sqrt{2}}$

(b) $2\sqrt{2}R$

(c) $\frac{R}{\sqrt{2}}$

(d) $\sqrt{2}R$

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2. If a point charge $q$ is placed at the centre of a cube, then the electric flux linked with each face of the cube is

(a) $\frac{q}{\varepsilon_0}$

(b) $\frac{q}{6\varepsilon_0}$

(c) $\frac{3q}{\varepsilon_0}$

(d) $\frac{q}{3\varepsilon_0}$

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3. In a region, a uniform electric field is given by $\vec{E} = (2\hat{i} + 6\hat{j} + 8\hat{k})$ N/C
The electric flux through a sphere of radius 1 m, centered at the origin is

(a) 8 N-m$^2$/C

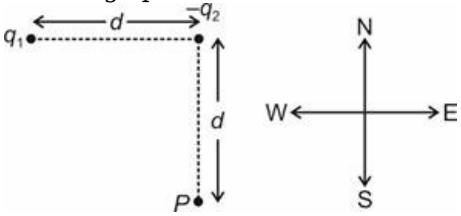
(b) Zero

(c) 10 N-m$^2$/C

(d) 16 N-m$^2$/C

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4. In space two charges are placed as shown in figure. What is possible direction of electric field at point $P$ out of the following option?




(a) South-West

(b) West

(c) North-East

(d) South

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5. A positive charge is moved from low potential point A to a high potential point B. Then the electric potential energy

(a) Increases

(b) Decreases

(c) Will remain same

(d) Nothing definite can be predicted

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6. A parallel plate capacitor is charged and then charging battery is disconnected. If the plates are now pulled apart with insulated handles then

(a) The capacitance increases

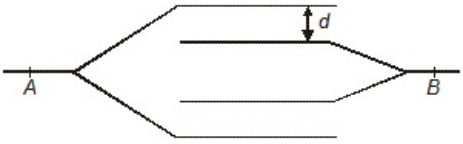
(b) Potential energy decreases

(c) Potential difference increases

(d) Charge and potential difference both remains same

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7. Four plates each of area $A$ at distance $d$ from each other are arranged as shown. The net capacitance of arrangement would be




(a) $\frac{3\varepsilon_0 A}{d}$

(b) $\frac{\varepsilon_0 A}{d}$

(c) $\frac{4\varepsilon_0 A}{d}$

(d) $\frac{2\varepsilon_0 A}{d}$

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8. Two metallic spheres $A$ and $B$ of radius '$a$' and '$b$' respectively are at same electric potential. The ratio of the surface charge densities of sphere $A$ and $B$ is

(a) $\frac{b^2}{a^2}$

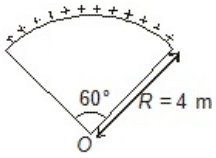
(b) $\frac{b}{a}$

(c) $\frac{a^2}{b^2}$

(d) $\frac{a}{b}$

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9. Find the electric potential at the point $O$ due to segment of ring as shown in figure, whose linear charge density is 4 C/m




(a) $3.8 \times 10^{12}$ V

(b) $3.8 \times 10^{10}$ V

(c) $4.8 \times 10^{10}$ V

(d) $4.8 \times 10^{12}$ V

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10. A 0.8 m long solenoid has 800 turns and has a field density of $2.52 \times 10^{-3}$ T at its centre. What is current in the wire?

(a) 3 A

(b) 2 A

(c) 1 A

(d) 4 A

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11. A magnet of magnetic moment $\vec{M} = (50\hat{i})\ A\ m^2$ is placed in a magnetic field $\vec{B} = (0.2\hat{i} + 0.3\hat{j})$ T . The torque acting on the magnet is

(a) $(15\hat{k})$ N m

(b) $(15\hat{k} + 10\hat{j})\ N\ m$

(c) $10\hat{k}\ N\ m$

(d) Zero

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12. A galvanometer of resistance 999 Ω gives full scale deflection for 1 mA. Resistance of the shunt required to convert it into an ammeter of range (0 to 1 A) is

(a) 1 Ω

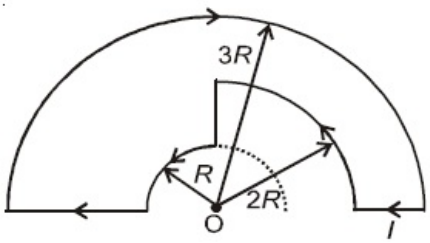
(b) 999 Ω

(c) 99.9 Ω

(d) 0.999 Ω

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13. Magnetic field at point $O$ is




(a) $\frac{\mu_0 I}{16R} \odot$

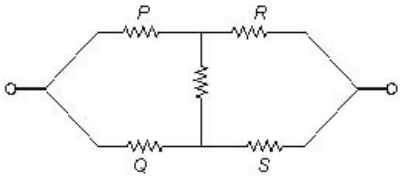
(b) $\frac{5\mu_0 I}{48R} \odot$

(c) $\frac{\mu_0 I}{12R} \otimes$

(d) $\frac{5\mu_0 I}{48R} \otimes$

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14. In Wheatstone's network, $P = 4\ \Omega$, $Q = 4\ \Omega$, $R = 4\ \Omega$ and $S = 6\ \Omega$. The resistance with which $S$ should be shunted in order that the bridge may be balanced is




(a) 4 Ω

(b) 6 Ω

(c) 8 Ω

(d) 12 Ω

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15. A bulb has voltage rating and power rating are 220 V and 110 W respectively. The resistance of the bulb is

(a) 110 Ω

(b) 220 Ω

(c) 330 Ω

(d) 440 Ω

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16. When $1.6 \times 10^{19}$ electrons are removed from a neutral metal plate, the electric charge on it will be

(a) 1 C

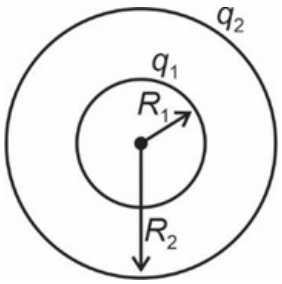
(b) –1 C

(c) 2.5 C

(d) –2.5 C

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17. Two uniformly charged conducting spherical shells are shown in figure. Both the shells are concentric and have radii $R_1$ and $R_2$ and $r$ is the distance from their common centre




| | | | |
|---|---|---|---|
| A. | Electric potential for $r < R_1$ | (P) | Zero |
| B. | Electric potential for $R_1 < r < R_2$ | (Q) | Non-zero constant |
| C. | Electric field for $r < R_1$ | (R) | Variable |
| D. | Electric field for $R_1 < r < R_2$ | (S) | Cannot be concluded |

(a) A(Q), B(R), C(P), D(S)

(b) A(Q), B(R), C(P), D(R)

(c) A(Q), B(P), C(P), D(R)

(d) A(Q), B(P), C(R), D(S)

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18. The intensity of electric field required to keep a proton of mass $1.6 \times 10^{-27}$ kg and charge $1.6 \times 10^{-19}$ C in

equilibrium is nearly

(a) $10^{7}$ V/m

(b) $10^{-5}$ V/m

(c) $10^{-19}$ V/m

(d) $10^{-7}$ V/m

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19. An electric dipole consisting of two equal and opposite charges of magnitude $3 \times 10^{-4}$ C separated by a distance of 1 cm is placed in an electric field of $4 \times 10^{6}$ N/C. The maximum torque that can be applied by the electric field on the dipole will be

(a) 1.2 N m

(b) 12 N m

(c) 6 N m

(d) Zero

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20. If $\phi_1$ is the magnitude of electric flux which is entering and $\phi_2$ is the magnitude of electric flux leaving an enclosed surface, then the electric charge inside the surface is

(a) $(\phi_1 - \phi_2)\varepsilon_0$

(b) $(\phi_2 - \phi_1)\varepsilon_0$

(c) $(\phi_1 + \phi_2)\varepsilon_0$

(d) $\frac{(\phi_2 - \phi_1)}{\varepsilon_0}$

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21. Electric potential in space varies as $V = -3x + 5y + \sqrt{15}z$, the magnitude of electric field is

(a) 7

(b) $7\sqrt{2}$

(c) 5

(d) $5\sqrt{2}$

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22. A capacitor stores energy $U$ when it has charge $Q$ on it. If the charge is increased to $2Q$, the stored energy will become

(a) $2U$

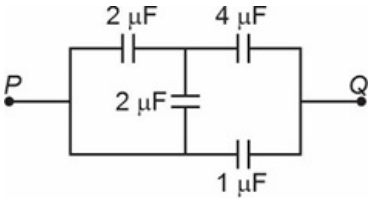
(b) $\frac{U}{4}$

(c) $16U$

(d) $4U$

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23. In the given circuit diagram, the resultant capacitance between $P$ and $Q$ is




(a) 6 μ F

(b) 3 μ F

(c) 9 μ F

(d) 2 μ F

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24. Three capacitors of capacities 1 μ F, 2 μ F and 3 μ F are connected in series and a potential difference of 22 V is applied across the combination of the capacitors. Then, the charge that will appear on the positive plate of 3 μ F capacitor will be

(a) 132 μ C

(b) 22 μ C

(c) 11 μ C

(d) 12 μ C

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25. If the distance between the plates of a parallel plate air capacitor is doubled and a dielectric of dielectric constant 2 is introduced between the plates, then the capacitance

(a) Remain same

(b) Will become double

(c) Will become quadruple

(d) Will become half

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26. Force of attraction between the plates of a charged parallel plate capacitor is (symbols have their usual meanings)

(a) $\frac{Q^2}{A\varepsilon_0}$

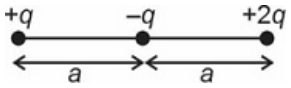
(b) $\frac{2Q^2}{A\varepsilon_0}$

(c) $\frac{Q^2}{2A\varepsilon_0}$

(d) $\frac{\sqrt{2}Q^2}{A\varepsilon_0}$

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27. Three point charges are placed in a line as shown in the figure. The potential energy of the system is




(a) $\frac{q^2}{2\pi\varepsilon_0 a}$

(b) $\frac{-q^2}{4\pi\varepsilon_0 a}$

(c) $\frac{-q^2}{\pi\varepsilon_0 a}$

(d) $\frac{-q^2}{2\pi\varepsilon_0 a}$

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28. In a conductor through a cross-section 4 C charge flows from left to right and –2 C charge flows from right to left in 2 second. The average value of electric current is

(a) 1 A

(b) 3 A

(c) 2 A

(d) 6 A

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29. The specific resistance or resistivity of a wire

(a) Decreases by increasing area of cross-section

(b) Decreases by increasing length of the wire

(c) Is independent of length and cross section of the wire

(d) Is same irrespective of the material of the wire

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30. The resistance of a wire is 20 Ω . If the length of the wire is increased by 10% by stretching it, then the new resistance of the wire will be

(a) 20 Ω

(b) 12.1 Ω

(c) 24.2 Ω

(d) 25.5 Ω

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31. A cell of emf 8 V is connected with an external resistance of 3 Ω , the potential difference across the cell is measured as 4 V. What is the value of internal resistance of the cell?

(a) 1 Ω

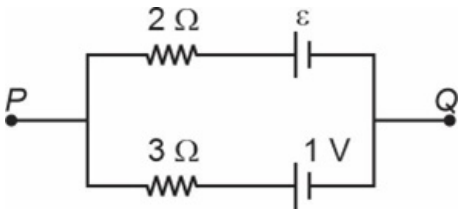
(b) 3 Ω

(c) $\frac{3}{2}$ Ω

(d) 4 Ω

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32. If potential difference between points $P$ and $Q$ is 2 V, then the value of ε in the figure is




(a) 4 V

(b) 8 V

(c) $\frac{4}{3}$ V

(d) $\frac{8}{3}$ V

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33. A galvanometer can be converted into a voltmeter by connecting

(a) High resistance in series with galvanometer

(b) Low resistance in series with galvanometer

(c) High resistance in parallel with galvanometer

(d) Low resistance in parallel with galvanometer

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34. **Assertion (A):** The thermal coefficient of resistance is negative for metals and positive for non-metals.

**Reason (R):** With increase in temperature free electrons increase in metals and decrease in non-metals.

In the light of the above statements, choose the correct answer from the options given below

(a) Both (A) and (R) are true and (R) is the correct explanation of (A)

(b) Both (A) and (R) are true and (R) is not the correct explanation of (A)

(c) (A) is true and (R) is false

(d) (A) is false and (R) is false

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35. The magnetic field at a point due to a long straight wire carrying a current $I$ is proportional to

(a) $I$

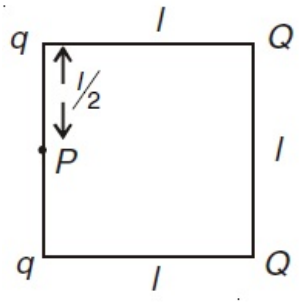
(b) $\sqrt{I}$

(c) $I^2$

(d) $I^{-1}$

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36. Charges $Q$ and $q$ are kept at corners of a square of side $l$ as shown in diagram. For what value of $q$ potential at point $P$ is zero?




(a) $q = \frac{Q}{\sqrt{5}}$

(b) $q = -\frac{Q}{\sqrt{3}}$

(c) $q = \frac{Q}{\sqrt{3}}$

(d) $q = -\frac{Q}{\sqrt{5}}$

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37. A capacitor is charged by using battery. Now a dielectric slab is introduced. What will happen to potential difference, if battery remains connected?

(a) Potential difference across plates increases

(b) Potential difference across plates decreases

(c) Potential difference across plates does not change

(d) Potential difference across plates becomes zero

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38. **Assertion (A) :** Electric force on the positive charge act in the direction of electric field.

**Reason (R) :** Electric field intensity due to a point charge will be different at different point.

In the light of above statements, select the correct option

(a) Both (A) and (R) are true, (R) correct explanation of (A)

(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A)

(c) (A) is true, (R) is false

(d) (A) is false, (R) is true

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39. A charge $q$ is placed at the centre of the line joining two equal charges $Q$. The system of the three charges will be in equilibrium, if $Q$ is equal to

(a) $-q$

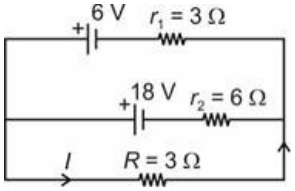
(b) $-2q$

(c) $-4q$

(d) $\frac{-q}{2}$

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40. Consider the following combination of cells $A$ and $B$ which are supplying currents.




To obtain the same current through $R = 3\ \Omega$ resistor, the circuit can be made up of a single cell of

(a) 10 V emf, internal resistance 2 Ω

(b) 10 V emf, internal resistance 0.5 Ω

(c) 9 V emf, internal resistance 2 Ω

(d) 12 V emf, internal resistance 2 Ω

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41. Two identical bulbs when connected in parallel to source take 60 watt each. The total power consumed, when they are connected in series with same source is

(a) 30 W

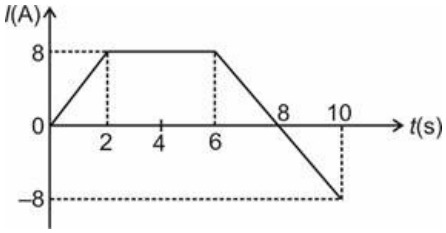
(b) 120 W

(c) 15 W

(d) 60 W

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42. The current $I$ through a conductor varies with time $t$ as shown in figure. The average electric current during $t = 0$ to $t = 10$ s will be




(a) 5.6 A

(b) 4 A

(c) 6.5 A

(d) 7.6 A

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43. The current through a wire depends on time as $I = (6t^2 + 6t)$ A. The charge flowing through cross-section of wire from $t = 0$ to $t = 3$ second is

(a) 60 C

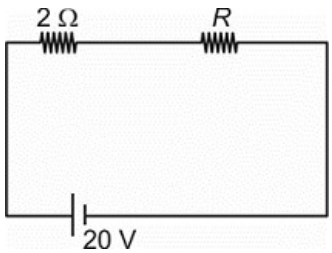
(b) 22 C

(c) 54 C

(d) 81 C

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44. If the power dissipation in the circuit shown in figure is 100 W, then value of $R$ is




(a) 1 Ω

(b) 2 Ω

(c) 4 Ω

(d) 3 Ω

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45. Two charged particles having charge 4 μC and mass $8 \times 10^{-3}$ kg each are joined by an insulating string of length 3 m and the system is kept on a smooth horizontal table. If suddenly string is cut then the acceleration of each particle is

(a) 2 $m/s^2$

(b) 4 $m/s^2$

(c) 6 $m/s^2$

(d) 8 $m/s^2$

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46. A proton and a deuteron enter into a uniform magnetic field perpendicularly. If both have same kinetic energy then the ratio of their radii of circular paths will be

(a) 1 : 2

(b) $1:\sqrt{2}$

(c) $3:\sqrt{2}$

(d) 1 : 4

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47. Two long parallel wires separated by distance $d$ carry current $i_1$ and $i_2$ in opposite directions, the force per unit length between them is

(a) $\frac{\mu_0 i_1 i_2}{2\pi d}$ , attractive

(b) $\frac{\mu_0 i_1 i_2}{2\pi d}$ , repulsive

(c) $\frac{2\mu_0 i_1 i_2}{\pi d}$ , attractive

(d) $\frac{\sqrt{2}\mu_0 i_1 i_2}{\pi d}$ , repulsive

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48. Consider the following statements.

A. Magnetic force on a moving charged particle is always non-zero.

B. Magnetic force can change kinetic energy of a charged particle.

C. Magnetic force can change linear momentum of a charged particle.

D. A charged particle at rest does not feel magnetic force on it.

The correct statement(s) is/are

(a) Only C

(b) A, B and C

(c) C and D

(d) B, C and D

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49. A proton moving with a velocity $6 \times 10^4$ m/s enters in a magnetic field of 0.5 tesla at an angle of 60° with the field. The radius of curvature of its path will be

(a) 1.2 cm

(b) 2.2 mm

(c) 1.0 mm

(d) 2.4 cm

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50. A current of 2 A is flowing along positive *x*-axis through a straight wire of length 1 m placed in a region of a magnetic field given by $\vec{B} = (\hat{i} + 2\hat{j})$ T . The magnitude of the force experienced by the wire is equal to

(a) 1 N

(b) 2 N

(c) 4 N

(d) 7 N

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51. The enthalpy of combustion of methane at 25°C is –890 kJ $mol^{-1}$. The heat liberated when 4.8 g of methane is burnt in air is

(a) 445 kJ

(b) 267 kJ

(c) 890 kJ

(d) 178 kJ

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52. For the reaction

$$CH_3COOH(l) + 2O_2(g) \rightleftharpoons 2CO_2(g) + 2H_2O(l)$$

At 25°C and 1 atm pressure, $\Delta H = -874$ kJ.

The change in internal energy ( $\Delta$ U) is

(a) –874 kJ

(b) –869 kJ

(c) –878.9 kJ

(d) 4874 kJ

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53. Which among the following match is incorrect?

| | $\Delta_r H°$ | $\Delta_r S°$ | $\Delta_r G°$ | **Description** |
|---|---|---|---|---|
| (1) | – | + | – | Reaction spontaneous at all temperature |
| (2) | – | – | – | Reaction spontaneous at low temperature |
| (3) | + | + | – | Reaction nonspontaneous at high temperature |
| (4) | + | – | + | Reaction nonspontaneous at all temperature |

(a) (1)

(b) (2)

(c) (3)

(d) (4)

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54. **Assertion (A):** The entropy of any pure crystalline substance approaches zero as the temperature approaches absolute zero.

**Reason (R):** In pure crystalline substances, there is perfect order in a crystal at absolute zero.

In the light of above statements, select the correct option

(a) Both (A) and (R) are true, (R) is the correct explanation of (A)

(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A)

(c) (A) is true, (R) is false

(d) (A) is false, (R) is true

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55. The value of $\Delta$ G° at 300 K if equilibrium constant for the reaction = $10^2$ is (R = gas constant)

(a) –300 R

(b) –2.303 × 600 R

(c) –2.303 × 300 R

(d) –600 R

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56. Given

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g), \ \Delta_r H° = -92.4 \text{ kJ mol}^{-1}$

The standard enthalpy of formation of $NH_3$ gas is

(a) $+46.2 \text{ kJ mol}^{-1}$

(b) $-92.4 \text{ kJ mol}^{-1}$

(c) $-46.2 \text{ kJ mol}^{-1}$

(d) $+92.4 \text{ kJ mol}^{-1}$

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57. The correct IUPAC name of $[CoCl_2(en)_2]Cl$ is

(a) Dichloridodiethane-1,2-diamine cobalt(III)chloride

(b) Bis(Ethane-1,2-dimanine) dichloridecobaltate(III)chloride

(c) Bis(ethane-1,2-diamine) dichloridecobalt(III)chloride

(d) Dichloridobis(ethane-1,2-diamine) cobalt(III)chloride

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58. The number of geometrical isomers of $[Cr(en)_2Cl_2]^+$ is

(a) Zero

(b) 2

(c) 3

(d) 4

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59. The correct increasing order of field strength of ligands is

(a) $NH_3 > H_2O > Cl^- > CN^-$

(b) $CN^- > NH_3 > H_2O > Cl^-$

(c) $Cl^- > CN^- > NH_3 > H_2O$

(d) $CN^- > H_2O > Cl^- > NH_3$

Pdf By: @Romlex69

60. Consider the following statements.

**Statement-I:** EDTA is used in the treatment of lead poisoning.

**Statement-II:** $[(Ph_3P)_3RhCl]$ is Wilkinson catalyst which is used for the hydrogenation of alkenes.

(a) Statement I is correct and II is incorrect

(b) Statement I is incorrect and II is correct

(c) Both statement I and II are correct

(d) Both statement I and II are incorrect

Pdf By: @Romlex69

61. The correct relation between $\Delta_t$ and $\Delta_0$ is

(a) $\Delta_t = \frac{4}{9}\Delta_o$

(b) $\Delta_t = \frac{2}{3}\Delta_o$

(c) $\Delta_o = 2\ \Delta_t$

(d) $\Delta_o = \frac{4}{9}\Delta_t$

Pdf By: @Romlex69

62. Crystal field stabilization energy for high spin $d^5$ octahedral complex is

(a) $-2\ \Delta_0$

(b) Zero

(c) $-1.2\ \Delta_0$

(d) $-0.6\ \Delta_0$

Pdf By: @Romlex69

63. The complexes $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5(SO_4)]Cl$ are the examples of which type of isomerism?

(a) Linkage isomerism

(b) Coordination isomerism

(c) Ionisation isomerism

(d) Solvate isomerism

Pdf By: @Romlex69

64. Which of the following carbonyls will have the strongest C – O bond?

(a) $[Mn(CO)_6]^+$

(b) $[V(CO)_6]^-$

(c) $[Fe(CO)_5]$

(d) $[Cr(CO)_6]$

Pdf By: @Romlex69

65. The inner orbital complex among the following is

(a) $[MnCl_6]^{3-}$

(b) $[Fe(H_2O)_6]^{3+}$

(c) $[Co(NH_3)_6]^{3+}$

(d) $[MnF_6]^{3-}$

Pdf By: @Romlex69

66. Among the following, the most stable complex is

(a) $[Fe(C_2O_4)_3]^{3-}$

(b) $[FeCl_6]^{3-}$

(c) $[Fe(H_2O)_6]^{3+}$

(d) $[Fe(NH_3)_6]^{3+}$

Pdf By: @Romlex69

67. The EAN of $Fe(CO)_5$ is

(a) 35

(b) 37

(c) 36

(d) 34

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68. Which of the following is a $\pi$ -acid ligand?

(a) $F^-$

(b) $H_2O$

(c) $NH_3$

(d) CO

Pdf By: @Romlex69

69. In solid $CuSO_4 \cdot 5H_2O$, number of $H_2O$ molecule coordinated to copper is equal to

(a) 5

(b) 2

(c) 3

(d) 4

Pdf By: @Romlex69

70. The correct geometry of $[Ni(CN)_4]^{2-}$ and $[NiCl_4]^{2-}$ respectively are

(a) Both are tetrahedral

(b) Square planar and tetrahedral

(c) Both are square planar

(d) Tetrahedral and square planer

Pdf By: @Romlex69

71. Unit of equilibrium constant $K_P$ for the reaction $CO_2(g) + C(s) \rightleftharpoons 2CO(g)$ is

(a) $(atm)^0$

(b) $(atm)^{-1}$

(c) $(atm)^1$

(d) $(atm)^2$

Pdf By: @Romlex69

72. Which of the following molecules acts as Lewis acid?

(a) $AlCl_3$

(b) $H_2O$

(c) $NH_3$

(d) $(CH_3)_3P$

Pdf By: @Romlex69

73. The pH of 0.01 M aq. NaA solution is ($K_a$ for HA is $1 \times 10^{-4}$)

(a) 6

(b) 12

(c) 10

(d) 8

Pdf By: @Romlex69

74. Consider the following reaction at equilibrium
$A(aq) + B(aq) \rightleftharpoons 3\ C\ (aq)$
If the equilibrium concentration of A, B and C are 2 M, 2 M and 4 M respectively then standard Gibbs free energy change for reaction will be

(a) –RT ln 2

(b) –4 RT ln2

(c) –3 RT ln2

(d) –8 RT ln2

Pdf By: @Romlex69

75. If the dissociation constant of 0.01 M $CH_3COOH$ is $10^{-5}$, then the dissociation constant of its conjugate base at 25°C is

(a) $10^{-11}$

(b) $10^{-9}$

(c) $10^{-8}$

(d) $10^{-12}$

Pdf By: @Romlex69

76. If equal volume of two solutions of pH 2 and 3 are mixed then pH of resulting solution is (log (5.5) = 0.74)

(a) 2.1

(b) 2.26

(c) 3.26

(d) 4.2

Pdf By: @Romlex69

77. When 200 ml of 0.2 M $CH_3CO_2H$ is mixed with 200 ml of 0.2 M NaOH then pH of resulting solution is [$pK_a$ for $CH_3CO_2H$ = 4.74 and log 2 = 0.3]

(a) 9.1

(b) 8.52

(c) 8.87

(d) 4.75

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78. The dissociation energy of $CH_4(g)$ is 360 kcalmol$^{-1}$ and that of $C_2H_6(g)$ is 620 kcalmol$^{-1}$. The C–C bond energy is

(a) 260 kcal/mol

(b) 180 kcal/mol

(c) 130 kcal/mol

(d) 80 kcal/mol

Pdf By: @Romlex69

79. The standard enthalpies of formation of $CO_2(g)$, $H_2O(l)$ and $C_2H_4(g)$ are –393.5, –241.8, +52.3 kJ/mole respectively. Then the standard enthalpy of combustion of ethylene gas is

(a) +1322.9 kJ/mole

(b) –1322.9 kJ/mole

(c) +1218.3 kJ/mole

(d) –1218.3 kJ/mole

Pdf By: @Romlex69

80. If for a reaction $A(g) \rightarrow B(g) + C(g)$, $\Delta H = 30$ kJ/mol and $\Delta S = 60$ J/Kmol, then temperature at which reaction will be spontaneous is

(a) 520 K

(b) 127 K

(c) 227 K

(d) 27 K

Pdf By: @Romlex69

81. Which of the following is not an intensive property?

(a) Enthalpy

(b) Temperature

(c) Pressure

(d) Melting point

Pdf By: @Romlex69

82. Given below are the two statements Assertion (A) and Reason (R)

**Assertion (A):** For adiabatic process, q = 0.

**Reason (R):** For adiabatic process work is done on the expense of change in internal energy.

In the light of above statements choose the correct format.

(a) Both (A) and (R) are true and the (R) is the correct explanation of the (A)

(b) Both (A) and (R) are true but the (R) is not the correct explanation of the (A)

(c) (A) is true statement but (R) is false

(d) Both (A) and (R) are false statements

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83. 2 mol of an ideal gas expanded isothermally and reversibly from 1 L to 10 L at 300 K. What is the enthalpy change?

(a) – 11.47 kJ

(b) 11.47 kJ

(c) 0 kJ

(d) 4.98 kJ

Pdf By: @Romlex69

84. $XY_2(g) \rightleftharpoons XY(g) + Y(g)$
Initial pressure of $XY_2$ is 600 mm Hg. The total pressure at equilibrium is 800 mm Hg. Assuming volume of system to remain contant, the value of $K_P$ is

(a) 50

(b) 100

(c) 20

(d) 400

Pdf By: @Romlex69

85. Solubility of $PbCl_2(s)$ is minimum in

(a) Water

(b) 0.1 M $Pb(NO_3)_2$

(c) 0.1 M $CaCl_2$

(d) 0.1 M $AlCl_3$

Pdf By: @Romlex69

86. The primary and secondary valences of Co metal ion in $[Co(NH_3)_4Cl_2]Cl$ respectively are

(a) 3 and 6

(b) 3 and 3

(c) 6 and 6

(d) 6 and 3

Pdf By: @Romlex69

87. The pair of ambidentate ligands among the following is

(a) $NH_3$, $OH^-$

(b) $NO_2^-$, $H_2O$

(c) $SCN^-$, $NO_2^-$

(d) $C_2O_4^{2-}$, $NH_3$

Pdf By: @Romlex69

88. Which of the following will have lowest molar conductance at infinite dilution?

(a) $[Pd(NH_3)_2Cl_4]$

(b) $[Pd(NH_3)_3Cl_3]Cl$

(c) $[Pd(NH_3)]_4Cl_2]Cl_2$

(d) $[Pd(NH_3)]_5Cl]Cl_3$

Pdf By: @Romlex69

89. The oxidation number of Cu in $[Cu(CN)_4]^{3-}$ is

(a) +2

(b) +1

(c) Zero

(d) +3

Pdf By: @Romlex69

90. Which among the following is a heteroleptic complex?

(a) $[Co(NH_3)_6]Cl_3$

(b) $[Co(SCN)_4]^{2-}$

(c) $[NiCl_2(PPh_3)_2]$

(d) $[Cr(H_2O)_6]^{3+}$

Pdf By: @Romlex69

91. For the reaction: $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ The forward reaction at constant temperature is favoured by

(a) Introducing $PCl_3(g)$ at constant volume

(b) Introducing inert gas at constant volume

(c) Introducing $PCl_5(g)$ at constant volume

(d) Introducing $Cl_2(g)$ at constant volume

Pdf By: @Romlex69

92. Which of the following equations correctly represents the standard enthalpy of formation ( $\Delta_f H°$) of methane?

(a) $C(diamond) + 4H(g) \rightarrow CH_4(g)$

(b) $C(graphite) + 2H_2(g) \rightarrow CH_4(g)$

(c) $C(diamond) + 2H_2(g) \rightarrow CH_4(g)$

(d) $C(graphite) + 4H(g) \rightarrow CH_4(g)$

Pdf By: @Romlex69

93. Introduction of inert gas (at same temperature) will affect the equilibrium if

(a) Volume is constant and $\Delta n_g \neq 0$

(b) Pressure is constant and $\Delta n_g \neq 0$

(c) Volume is constant and $\Delta n_g = 0$

(d) Pressure is constant and $\Delta n_g = 0$

Pdf By: @Romlex69

94. The reaction for which, $K_p = K_c$ is satisfied for

I. $A(g) + 2B(g) \rightleftharpoons 3C(g)$

II. $A(s) \rightleftharpoons B(g)$

III. $2A(g) \rightleftharpoons B(g) + C(g)$

IV. $A(s) + B(g) \rightleftharpoons C(s) + 2(s) + 2D(g)$

(a) I and II only

(b) II and III only

(c) I and III only

(d) I, II and IV only

Pdf By: @Romlex69

95. The pH of $\frac{M}{100}$ NaOH is

(a) 2

(b) 4

(c) 12

(d) 10

Pdf By: @Romlex69

96. In the following question, a statement of assertion **(A)** is followed by a statement of reason **(R)**

**Assertion (A):** pH of salt solution of weak acid and weak base is independent of concentration of solution.

**Reason (R):** Degree of hydrolysis of salt solution of weak acid and weak base is directly proportional to the concentration of the salt.

(a) Both (A) and (R) are true and the (R) is the correct explanation of the (A).

(b) Both (A) and (R) are true but the (R) is not the correct explanation of the (A).

(c) (A) is true statement but (R) is false.

(d) Both (A) and (R) are false statements.

Pdf By: @Romlex69

97. For strong acid strong base neutralisation energy for 1 mole $H_2O$ formation is –57.1 kJ. If 0.25 mole of strong monoprotic acid is reacted with 0.5 mole of strong base then enthalpy of neutralisation in kJ is

(a) $-(0.25 \times 57.1)$

(b) $0.5 \times 57.1$

(c) 57.1

(d) $-(0.5 \times 57.1)$

Pdf By: @Romlex69

98. From the following bond energies

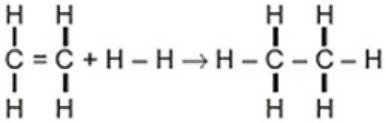
H – H bond energy : 431.37 kJ $mol^{-1}$

C = C bond energy : 606.10 kJ $mol^{-1}$

C – C bond energy : 336.49 kJ $mol^{-1}$

C – H bond energy : 410.50 kJ $mol^{-1}$

Enthalpy for the reaction,



will be

(a) 553.0 kJ $mol^{-1}$

(b) 1523.6kJ $mol^{-1}$

(c) 243.6 kJ $mol^{-1}$

(d) 120.0 kJ $mol^{-1}$

Pdf By: @Romlex69

99. If the enthalpy change for the transition of a liquid to its vapour is 60 kJ $mol^{-1}$ at 300 K, then the entropy change for the process is

(a) 150 J $K^{-1}$ $mol^{-1}$

(b) 250 J $K^{-1}$ $mol^{-1}$

(c) 200 J $K^{-1}$ $mol^{-1}$

(d) 100 J $K^{-1}$ $mol^{-1}$

Pdf By: @Romlex69

100. For the reaction

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$K_p$ is equal to

(a) $K_C$

(b) $K_C RT$

(c) $K_C(RT)^2$

(d) $K_C(RT)^{-1}$

Pdf By: @Romlex69

101. The term gamosepalous is used for

(a) Corolla when its components are free

(b) The components of corolla attached with gynoecium

(c) Sepals which are united with gynoecium

(d) Calyx when its components are united

Pdf By: @Romlex69

102. The last step of DNA fingerprinting is

(a) Blotting

(b) Hybridisation

(c) Electrophoresis

(d) Autoradiography

Pdf By: @Romlex69

103. Select the **incorrect** match.

(a) Gene - Codes for rRNA or tRNA

(b) Cistron - Segment of DNA codes for polypeptide

(c) Exons - Expressed sequences

(d) Introns - Appear in mature mRNA

Pdf By: @Romlex69

104. Mark the **incorrect** statement w.r.t. features of human genome.

(a) The human genome contains 3164.7 million bp

(b) The average gene consists of 3000 bases

(c) Less than 2 percent of genome codes for proteins

(d) Chromosome 1 has fewest genes

Pdf By: @Romlex69

105. The genetic codon for which no tRNA is found is

(a) AUG

(b) UUU

(c) UUA

(d) UAG

Pdf By: @Romlex69

106. The packaging of chromatin in eukaryotes at higher level requires additional set of proteins that are collectively referred to as

(a) Histone proteins

(b) Nucleosomes

(c) Non-Histone Chromosomal (NHC) proteins

(d) Chromatins

Pdf By: @Romlex69

107. Which of the given acts as a ribozyme during translation in prokaryotes?

(a) Reverse transcriptase

(b) 18S rRNA

(c) 23S rRNA

(d) 5S rRNA

Pdf By: @Romlex69

108. BOD is

(a) A measure of the organic matter present in the water

(b) Significantly reduced in polluted water

(c) Amount of nitrogen in water

(d) Measure of the rate of uptake of $CO_2$ by micro-organisms

Pdf By: @Romlex69

109. In sewage treatment, activated sludge

(a) Serves as inoculum in aeration tank

(b) Is produced during primary treatment

(c) Contains soil & small pebbles

(d) Is digested during tertiary treatment

Pdf By: @Romlex69

110. ____ serve as an important biofertilizer in paddy fields.

(a) Fungi

(b) Mycoplasma

(c) Lichens

(d) Cyanobacteria

Pdf By: @Romlex69

111. Statins have been commercialised as A and are produced by a B .

Select the **correct** option to fill in the blanks A and B.

(a) A – Blood-cholesterol lowering agents
B – Fungus

(b) A – Clot busters
B – Fungus

(c) A – Clot busters
B – Bacterium

(d) A – Immunosuppressive agents
B – Bacterium

Pdf By: @Romlex69

112. Identify the feature which is **not** true for mango fruit.

(a) Develops from monocarpellary ovary

(b) Mostly one seeded

(c) Has fleshy, juicy mesocarp

(d) Endocarp is fibrous edible part

Pdf By: @Romlex69

113. Perigynous condition is seen in

(a) Guava

(b) Cucumber

(c) Plum

(d) Brinjal

Pdf By: @Romlex69

114. An operon is

(a) A protein that binds to a RNA segment and blocks the attachment of DNA polymerase, halting transcription

(b) A transcriptionally regulated system found in all eukaryotes

(c) A reversibly-bound protein that enhances the transcription of particular segment of RNA

(d) A unit of DNA containing a cluster of genes under control of a promoter and a repressor

Pdf By: @Romlex69

115. “Dominance of RNA world” can be attributed to

(a) Presence of introns only

(b) The process of splicing

(c) The tailing process

(d) The process of capping

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116. Large holed swiss cheese is ripened by a

(a) Fungus

(b) Virus

(c) Protist

(d) Bacterium

Pdf By: @Romlex69

117. During fermentation of dough, cheese making and production of beverages, the main gas produced is

(a) $CH_4$

(b) $H_2S$

(c) $CO_2$

(d) $H_2$

Pdf By: @Romlex69

118. In capping,_____ is added to 5′ end of _____

(a) Unusual nucleotide, rRNA

(b) Methyl guanosine triphosphate, hnRNA

(c) Guanosine monophosphate, hnRNA

(d) Adenine residue, tRNA.

Pdf By: @Romlex69

119. There are evidences that _____was the first genetic material.

Select the **correct** option to fill in the blank.

(a) DNA

(b) RNA

(c) Nucleoid

(d) Protein

Pdf By: @Romlex69

120. Mark these statements true(**T**) or false(**F**) and select the **correct** option.

**Statement-A:** Sometimes regulatory sequences are defined as regulatory genes even though they do not code for any RNA or proteins.

**Statement-B:** DNA is preferred for transmission of genetic information over RNA.

**Statement-C:** SNPs are helpful to find the chromosomal locations for disease associated sequences.

| | **A** | **B** | **C** |
|---|---|---|---|
| (1) | T | F | F |
| (2) | T | F | T |
| (3) | F | T | T |
| (4) | T | T | T |

(a) (1)

(b) (2)

(c) (3)

(d) (4)

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121. Regarding double helix model of DNA, which of the following statements is **correct**?

(a) Backbone is formed by sugar-nitrogen base

(b) The two strands of polynucleotides are perpendicular to each other

(c) There are two hydrogen bonds between all base pairs

(d) Number of hydrogen bonds between guanine and cytosine is three

Pdf By: @Romlex69

122. Select the **incorrect** statement regarding baculoviruses.

(a) These viruses are species-specific

(b) Majority of them (used as biocontrol agents) are in the genus *Nucleopolyhedrovirus*.

(c) Have no negative impacts on non-target insects

(d) Broad spectrum insecticidal application

Pdf By: @Romlex69

123. Match the following columns and select the **correct** option.

| **Column I** | **Column II** |
|---|---|
| a. Cyclosporin A | (i) *Monascus purpureus* |
| b. Statins | (ii) *Lactobacillus* |
| c. Streptokinase | (iii) *Trichoderma polysporum* |

d. Lactic acid (iv) *Streptococcus*

(a) a(i), b(iv), c(iii), d(ii)

(b) a(iii), b(i), c(iv), d(ii)

(c) a(iii), b(i), c(ii), d(iv)

(d) a(i), b(iii), c(iv), d(ii)

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124. Which of the following features does **not** belong to mustard flowers?

(a) Polyadelphous stamens

(b) Tetramerous flower

(c) Ovary superior

(d) Actinomorphic flower

Pdf By: @Romlex69

125. Read the given Assertion (A) and Reason (R) statements and select the **correct** option.
**Assertion (A) :** Banana is said to be a parthenocarpic fruit.
**Reason (R) :** In banana ovary grows into fruit without fertilization.

(a) Both (A) and (R) are true and (R) is the correct explanation of the (A).

(b) Both (A) and (R) are true but the (R) is not the correct explanation of the (A).

(c) (A) is true statement but (R) is false.

(d) Both (A) and (R) are false statements.

Pdf By: @Romlex69

126. In monocotyledonous plants, the primary root is __(A)__ lived and is replaced by large number of __(B)__

Select the **correct** option for A and B.

(a) A-Short, B-Tap root

(b) A-Long, B-Fibrous root

(c) A-Short, B-Fibrous root

(d) A-Long, B-Tap root

Pdf By: @Romlex69

127. Choose the **incorrect** match among the following

(a) Marginal placentation – Pea

(b) Parietal placentation – Lemon

(c) Free central placentation – *Dianthus*

(d) Basal placentation – Sunflower

Pdf By: @Romlex69

128. The structure that holds the leaf blade to sunlight is

(a) Stipule

(b) Pulvinus

(c) Midrib

(d) Petiole

Pdf By: @Romlex69

129. China rose does **not** show

(a) Multilocular ovary

(b) Twisted aestivation of sepals

(c) Alternate type of phyllotaxy

(d) Stamens united into two bundles

Pdf By: @Romlex69

130. In roots, the cells proximal to region of meristematic activity

(a) Are very small with dense cytoplasm

(b) Are responsible for growth of the roots in the length

(c) Form very fine and delicate thread like structures called root hairs

(d) Is called region of maturation

Pdf By: @Romlex69

131. Sweet potato is modification of

(a) Stem

(b) Rhizome

(c) Adventitious roots

(d) Tap root

Pdf By: @Romlex69

132. Read the below given assertion (A) and reason (R) statements and select the **correct** option.

**Assertion (A):** RNA polymerase binds the promoter and initiates transcription.

**Reason (R):** RNA polymerase is only capable of catalysing the process of elongation.

(a) Both (A) and (R) are true statements and (R) is correct explanation of (A)

(b) Both (A) and (R) are true statements and (R) is not correct explanation of (A)

(c) (A) is true while (R) is false

(d) Both (A) and (R) are false

Pdf By: @Romlex69

133. A molecule that can act as genetic material must fulfill the following criteria, **except**

(a) It should be chemically and structurally stable

(b) It should be able to express itself in the form of Mendelian characters

(c) It should be able to generate its replica

(d) It should provide the scope for rapid mutation that are required for evolution

Pdf By: @Romlex69

134. Pappus is found in

(a) *Triticum*

(b) *Gossypium*

(c) *Hibiscus*

(d) *Helianthus*

Pdf By: @Romlex69

135. Read the following statements and choose the **correct** option.

**Statement-A:** Organic farmers work to create a system which will keep pest at manageable level by complex system of checks and balances within vibrant ecosystem.

**Statement-B:** Biocontrol refers to the use of biofertilizer that will enrich the nutrient quality of the soil.

(a) Only statement A is correct

(b) Only statement B is correct

(c) Both statements A and B are correct

(d) Both statements A and B are incorrect

Pdf By: @Romlex69

136. Ray florets of sunflower do **not** have

(a) Inferior ovary

(b) Pentacarpellary ovary

(c) Basal placentation

(d) United corolla

Pdf By: @Romlex69

137. Underground stem of zaminkand and *Colocasia* are modified to

(a) Perform photosynthesis

(b) Respire

(c) Protect the plant

(d) Store the food

Pdf By: @Romlex69

138. In *E. coli* which of the following occurs when glucose is low and lactose is abundant?

(a) Repressor binds to operator region

(b) β galactosidase breaks galactose into lactose and glucose

(c) Transport of lactose into the cell is enhanced

(d) RNA polymerase binds to operator

Pdf By: @Romlex69

139. Match the following column-I with column-II and choose the **correct** option.

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| a. | Mustard | (i) | $Epi_{5-7}\oplus\varphi\delta K_{(5)}\overset{\frown}{C_5A_{(\infty)}}\underline{G}_{(5)}$ |
| b. | Groundnut | (ii) | $Br\oplus\varphi\delta\overset{\frown}{P_{(3+3)}}A_{3+3}\underline{G}_{(3)}$ |
| c. | China rose | (iii) | $\oplus\varphi\delta K_{2+2}C_4A_{2+4}\underline{G}_{(2)}$ |
| d. | *Allium cepa* | (iv) | $\%\varphi\delta K_{(5)}C_{1+2+(2)}A_{(9)+1}\underline{G}_1$ |

(a) a(iv), b(i), c(ii), d(iii)

(b) a(iii), b(i), c(iv), d(ii)

(c) a(iv), b(iii), c(i), d(ii)

(d) a(iii), b(iv), c(i), d(ii)

Pdf By: @Romlex69

140. All the given flowers can be divided into two similar halves only in one particular vertical plane, **except**

(a) Gulmohur

(b) *Datura*

(c) Bean

(d) Pea

Pdf By: @Romlex69

141. Heterochromatin is

(a) Darkly stained region

(b) Loosely packed chromatin

(c) Transcriptionally inactive

The **correct** one(s) is/are

(a) Only (a)

(b) Only (a) and (b)

(c) Only (b) and (c)

(d) Only (a) and (c)

Pdf By: @Romlex69

142. Consider the following template sequence in DNA

3 ' TACGCAAATGTCGATCATATC 5 '

Choose the **incorrect** option for mRNA synthesised from the given sequence.

(a) Stop codon is present after fifth adenine in the RNA sequence

(b) Deletion of any one base prior to fifth adenine can change the amino acid number in the polypeptide synthesised

(c) Polypeptide synthesised from the mRNA will have only five amino acids

(d) There are no tRNA for the stop codon

Pdf By: @Romlex69

143. All of the following are the part of transcription unit in DNA, **except**

(a) Structural gene

(b) Promoter

(c) Repressor

(d) Terminator

Pdf By: @Romlex69

144. The untranslated region in mRNA

(a) Is the part of translational unit in mRNA

(b) Is required for efficient translation

(c) Is present after start codon and before stop codon

(d) Undergoes translation along with translational unit

Pdf By: @Romlex69

145. Choose the **incorrect** match among the following

(a) *z* gene – It codes for beta galactosidase for hydrolysis of disaccharide, lactose

(b) *i* gene – It synthesises repressor constitutively

(c) *a* gene – It encodes for transacetylase

(d) *y* gene – Encodes permease, which inhibits permeability of the cell to β -galactoside

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146. Floral diagram of Makoi plant does **not** represent

(a) Aestivation of petals

(b) Position of ovary with respect to other floral parts

(c) Adhesion of floral parts

(d) Number of floral parts

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147. In order to save major rivers of our country from water pollution, Ganga Action Plan and Yamuna Action Plan has been initiated by

(a) Ministry of Environment and Forests

(b) Khadi and Village Industries Commission

(c) Indian Agricultural Research Institute

(d) Forest Research Institute

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148. Citric acid is

(a) Produced by a protozoan

(b) Produced by *Acetobacter aceti*

(c) Produced by an eukaryotic organism

(d) Mainly used in detergent formulations

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149. Choose the **incorrect** match

(a) Dragonflies – Control aphids

(b) *Trichoderma* – Free living fungi

(c) Baculoviruses – Attack insects and other arthropods

(d) *Glomus* – Form mycorrhizal association

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150. During which treatment of the sewage, the suspended solids are removed?

(a) Biological treatment

(b) Secondary treatment

(c) Primary treatment

(d) Tertiary treatment

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151. If the cardiac output of a person is about 6.3 litres/minute and his stroke volume is 70 mL, then heart rate of the person will be

(a) 100/minute

(b) 90/minute

(c) 80/minute

(d) 70/minute

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152. Blood protein that mainly helps in maintaining osmotic balance is

(a) Fibrinogen

(b) Globulin

(c) Albumin

(d) Prothrombin

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153. Match the following columns w.r.t. human heart valves and their location.

| | Column-I | | Column-II |
|---|---|---|---|
| a. | Tricuspid valve | (i) | Aorta and pulmonary trunk |
| b. | Mitral valve | (ii) | Between left atrium and left ventricle |
| c. | Semilunar valves | (iii) | Between right atrium and right ventricle |

Choose the **correct** option.

(a) a(ii), b(i), c(iii)

(b) a(iii), b(i), c(ii)

(c) a(ii), b(iii), c(i)

(d) a(iii), b(ii), c(i)

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154. Each kidney of an adult human measures __A__ cm in length and __B__ cm in width.

Choose the option that **correctly** fills the blanks A and B respectively

(a) 5-7, 10-12

(b) 2-3, 5-7

(c) 10-12, 5-7

(d) 10-12, 2-3

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155. Counter current mechanism between loop of Henle and vasa recta

(a) Produces and maintain interstitial concentration gradient from cortex to inner medulla.

(b) Results in reabsorption of water in aescending limb of loop of Henle and collecting duct.

(c) Results in reabsorption of water in distal convoluted tubule and PCT.

(d) Produces osmolarity gradient 1200 mOsmol $L^{-1}$ to 300 mOsmol $L^{-1}$ from outer cortex to inner medulla.

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156. Which of the following **cannot** be considered as a difference between cortical and juxtamedullary nephrons?

(a) Length of loop of Henle

(b) Presence or absence of vasa recta

(c) Presence or absence of renal corpuscles

(d) Degree of extension of loop of Henle into the medulla

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157. Which of the following animals has the least concentrated urine relative to its blood plasma?

(a) Birds

(b) Human

(c) Fresh water fishes

(d) Camel

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158. In humans, right kidney is located lower in the abdominal cavity than the left one, because of more space being occupied by the

(a) Spleen

(b) Liver

(c) Stomach

(d) Pancreas

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159. The filtrate in which of the following part(s) of a nephron in humans has osmolarity less than 300 $mOsmolL^{-1}$?

(a) PCT and collecting duct

(b) PCT only

(c) Loop of Henle

(d) DCT

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160. In humans, a comparison of the volume of the glomerular filtrate formed per day with that of the urine released per day suggest that approximately __X_____ % of the filtrate is reabsorbed by the renal tubules. Here 'X' is

(a) 99

(b) 39

(c) 19

(d) 59

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161. Select the **incorrectly** matched option.

(1) Red blood cells – Help in transport of respiratory gases

(2) Neutrophils – Destroy foreign organisms entering the body
(3) Basophils – Produce antibodies
(4) Thrombocytes – Involve in coagulation of blood

(a) (1)

(b) (2)

(c) (3)

(d) (4)

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162. Which of the following statements is not **incorrect** w.r.t. cardiac cycle in a healthy human under normal physiological conditions?

(a) Ventricular systole increases the auricular pressure causing the closure of tricuspid and bicuspid valve.

(b) During ventricular diastole, semilunar valves are closed

(c) The second heart sound is associated with the closure of the tricuspid and bicuspid valves.

(d) The heart sounds are not of any clinical diagnostic significance.

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163. Neural signals through the sympathetic nerves can increase all of the following, **except**

(a) Heart rate

(b) Duration of cardiac cycle

(c) Cardiac output

(d) Speed of conduction of action potential

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164. Patients 'X' and 'Y' are hospitalised. The heart of 'X' is not pumping blood effectively enough to meet the body's need while the heart muscles in 'Y' is unexpectedly damaged by insufficient blood supply.

Identify the diseases 'X' and 'Y' are suffering from respectively and choose the **correct** option.

(a) Cardiac arrest, heart failure

(b) Heart failure, heart attack

(c) Heart attack, angina pectoris

(d) Angina pectoris, heart failure

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165. In animals, the circulatory pattern 'P' is considered to be more advantageous than the circulatory pattern 'Q'. The flow of fluid in former can be precisely regulated. 'P' is usually found in

(a) Arthropods

(b) Fishes

(c) Most of molluscs

(d) Echinoderms

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166. When glomerular blood pressure/glomerular blood flow drops, specific cells called 'X' become activated and release the enzyme 'Y', which converts angiotensinogen into a potent vasoconstrictor 'Z'. Identify X, Y, Z and choose the **correct** option.

(1) X → Rennin – Y → JG cells
(2) Y → Aldosterone – Z → Angiotensin II
(3) X → JG cells – Y → Renin
(4) X → JG cells – Y → Rennin

(a) (1)

(b) (2)

(c) (3)

(d) (4)

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167. In humans, which of the following are responsible for maintaining the increasing osmolarity gradient in the medullary interstitium of kidney?

(a) $H_2O$ and NaCl

(b) $H_2O$ and urea

(c) NaCl and $HCO_3^-$

(d) NaCl and urea

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168. All of the following structures of nephrons are completely situated in cortical region of kidneys, **except**

(a) Malpighian corpuscle

(b) Proximal convoluted tubule

(c) Distal convoluted tubule

(d) Loop of Henle

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169. Select the option among the following that **correctly** explains the function of a specific part of human nephron?

(1) Collecting duct – Maximum absorption of $H^+$ and $NH_3$
(2) Proximal convoluted tubules – Allow passage of small amount of urea into the medullary interstitium

(3) Distal convoluted tubules – Selective secretion of $H^+$, $K^+$ and $NH_3$

(4) Loop of Henle – Conditional reabsorption of $Na^+$ and water

(a) (1)

(b) (2)

(c) (3)

(d) (4)

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170. The hepatic portal vein carries blood from 'a' to 'b' before it is delivered to the systemic circulation.

Select the option that **correctly** identifies 'a' and 'b' respectively

(a) Intestine, kidney

(b) Kidney, intestine

(c) Liver, intestine

(d) Intestine, liver

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171. Which of the following options gives us the **correct** representation of organisms w.r.t. their nitrogenous wastes?

| | **Ammonotelic** | **Ureotelic** | **Uricotelic** |
|---|---|---|---|
| (1) | Reptiles | Birds | Land snails |
| (2) | Aquatic insects | Frog | Insects |
| (3) | Marine fishes | Reptiles | Mammals |
| (4) | Terrestrial amphibian | Marine fishes | Aquatic amphibians |

(a) (1)

(b) (2)

(c) (3)

(d) (4)

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172. In humans, specialised cardiac musculature responsible for generating the maximum number of action potentials is present in

(a) Right upper corner of the right atrium

(b) Left upper corner of the left atrium

(c) Lower left corner of the right atrium

(d) Right upper corner of the right ventricle

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173. During blood coagulation, the conversion of fibrinogen to its active form is caused by an enzyme 'E'. Select the option that represents the inactive form of 'E'.

(a) Thrombokinase

(b) Prothrombin

(c) Thrombin

(d) Fibrin

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174. In a standard E.C.G. of a healthy individual, return of the ventricles from excited to normal state is represented by

(a) P-wave

(b) QR segment

(c) T-wave

(d) End of P-wave

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175. Read the following statements and select the **incorrect** one w.r.t. a healthy adult human.

(a) The volume of blood filtered by the kidneys per minute, constitutes about $1/5^{th}$ of the cardiac output

(b) 180 L filtrate is formed by kidneys per day in a healthy adult human.

(c) On an average, 25-30 gm of urea is excreted out per day.

(d) Our lungs remove approximately 2000 mL of $CO_2$ everyday.

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176. Which of the following would be described as a portal system?

(a) An area connecting arterioles to venules

(b) A series of vessels that returns blood to the heart in an animal with an open circulatory system

(c) A space within or between organs where blood is allowed to pool

(d) A vessel or vessels connecting two capillary bed

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177. In humans, the per cent concentration of proteins in the blood plasma is equal to the per cent concentration of

(a) Haemoglobin in the blood plasma

(b) Monocytes in total WBCs

(c) Leucocytes in total WBCs

(d) Formed elements in the blood

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178. Consider the following for the most abundant cells of the blood in humans under normal conditions.

a. Formed in red bone marrow in adults

b. Biconvex in shape

c. Nucleus is bilobed

d. Usually destroyed in spleen

Select the **correct** option.

(a) a and b

(b) b and c

(c) c and d

(d) a and d

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179. Select the **incorrect** statement.

(a) Lymph has the same mineral distribution as that of plasma.

(b) Lymph is an important carrier for nutrients, hormones *etc*.

(c) The pumping organ of human body is mesodermally derived.

(d) Chordae tendinae prevents the backflow of blood from vena cava to left atrium.

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180. Which of the following events does not occur during the joint diastole?

(a) Mitral valves are open

(b) Blood flows from atria to ventricles

(c) Aortic valves are open

(d) Blood enters into right atrium through vena cava

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181. All of the following are true for the excretory products of larvae of frogs and *Labeo,* **except**

(a) Highly soluble in water

(b) More toxic than the excretory product of adult *Rana*

(c) Requires minimum amount of water for its elimination

(d) Also excreted by crustaceans

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182. How many of the substances given in the box below are actively reabsorbed in the renal tubules?

| Glucose, $Na^+$, Urea, $H_2O$, Amino acids, $H^+$ |
|---|

Select the **correct** option.

(a) Four

(b) Three

(c) Five

(d) One

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183. What will be the effect of a drug 'X' that blocks the transport of $Na^+$ and $Cl^-$ in the ascending limb of the loop of Henle.?

(a) Increase in the volume of urine

(b) No significant change in the urine volume

(c) There will be no micturition

(d) Decrease in the volume of urine

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184. 'A' is an anticoagulant used during the process of haemodialysis. 'A' is also produced within the human body by

a. Basophils

b. Monocytes

c. Mast cells

d. Eosinophils

Choose the **correct** option.

(a) a and b

(b) b and c

(c) a and c

(d) a and d

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185. Select the **incorrect** statement w.r.t. humans.

(a) Bile pigments secreted by a secondary lymphoid organ are excreted *via* digestive wastes.

(b) Small amounts of nitrogenous wastes are excreted *via* saliva too.

(c) The primary function of sweat is to facilitate a cooling effect on the body surface.

(d) Lungs remove a significant amount of water everyday.

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186. The “columns of Bertini” in the kidneys of humans arise as extensions of

(a) Cortex into medulla

(b) Cortex into renal pelvis

(c) Medulla into cortex

(d) Medulla into renal pelvis

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187. An important secretory constituent of sebaceous glands is/are

(a) Sterols

(b) Urea

(c) Glucose

(d) Amino acids

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188. Glomerular filtration occurs through filtration membranes which contains all, **except**

(a) Epithelial cells of Bowman’s capsule

(b) Endothelium of glomerular blood capillaries

(c) Epithelial cells of efferent arteriole

(d) Basement membrane

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189. All of the following factors favour the formation of concentrated urine, **except**

(a) Suppression of release of vasopressin

(b) Activation of osmoreceptors in the body

(c) Release of aldosterone from adrenal cortex

(d) Increased osmolarity towards the inner medullary interstitium

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190. What would be the effect on human heart if we give an injection of adrenaline to an individual?

(a) Increase in the rate of heart beat

(b) Decrease in the rate of heart beat

(c) Increase in the rate of heart beat but decrease in the strength of ventricular contraction

(d) Decrease in cardiac output

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191. The condition called erythroblastosis foetalis can be avoided by

(a) Injecting Rh antigen into mother's blood immediately after delivery of the first child

(b) Administering anti-Rh-antibodies to the newborn

(c) Administering anti-Rh antibodies to mother's blood immediately after delivery of first Rh positive child

(d) Injecting Rh positive blood group into mother's blood just after conception.

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192. What type of blood contains anti B-antibodies in the plasma and lacks Rh antigens?

(a) AB negative

(b) A negative

(c) B negative

(d) O positive

Pdf By: @Romlex69

193. Single circulation and venous heart is present in most

(a) Reptiles

(b) Fishes

(c) Birds

(d) Mammals

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194. **Assertion (A):** In humans, the myocardium is thickest in the wall of left ventricle.

**Reason (R) :** The left ventricle has to pump blood into all body organs, so bearing high work load.

In the light of above statements, select the **correct** option.

(a) (A) and (R) are true and the (R) is the correct explanation of the (A)

(b) (A) and (R) are true but the (R) is not the correct explanation of the (A)

(c) (A) is true statement but (R) is false

(d) (A) and (R) are false statements

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195. In humans, Malpighian body consists of

(a) Bowman's capsule with peritubular capillaries

(b) Only glomerulus with Juxtaglomerular apparatus

(c) Glomerulus with Bowman's capsule

(d) Bowman's capsule with collecting duct

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196. The **correct** pair of agranulocytes among the following is

(a) Monocytes and Basophils

(b) Monocytes and Lymphocytes

(c) Neutrophils and Lymphocytes

(d) Eosinophils and Lymphocytes

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197. In a blood transfusion case to avoid clumping of blood in a patient having blood group A, the donor blood group should be

(a) A and AB

(b) A, AB and B

(c) A and O

(d) A, B and O

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198. Closed circulatory system with 4 chambered heart is found in all of the following, **except**

(a) Birds

(b) Mammals

(c) Amphibians

(d) Crocodiles

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199. In humans, blood flow from the vena cava to the right atrium of heart due to

(a) Pressure difference between the vena cava and atrium

(b) Stimulation of SA-node generating electrical activity

(c) Stimulation of AV-node by the SA node

(d) Contraction of cardiac muscles

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200. Read the following statements w.r.t. human kidneys and choose the option with only **correct** statements.
a. The outer layer of kidney is a tough capsule.
b. Capsule divides kidney into outer cortex and inner medulla.
c. Medulla is divided into medullary pyramids by invagination of cortex into medulla.
d. Calyces project into medullary pyramids.

(a) a and b

(b) b and c

(c) c and d

(d) a and c

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